

MTH302 – Business Mathematics & Statistics
Spring 2022 – Quiz 3 (Solved)

Solved By:

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REMEMBER IN PRAYERS
KEEP SMILING
SAY NO TO CURRUPTION
HELP TO OTHERS

Regards,
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If A and B are mutually exclusive events then,

Select the correct option

☐ $P(A \text{ or } B) = P(A) - P(B)$

☒ $P(A \text{ or } B) = P(A) + P(B)$

☐ $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$

☐ $P(A \text{ and } B) = P(A) + P(B) + P(A \text{ or } B)$

Question # 8 of 10 (Start time: 07:42:07 PM, 30 August 2022)

An arrangement of data by successive time periods is called

Select the correct option



Representative of data



Time Series



Exponential smoothing



Linear regression



If an event A is a proper subset of an event B then Probability of the event

Select the correct option



None of these



A is equals the event B



B is less than the event A



A is less than the event B

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Question # 8 of 10 (start time: 05:13:06 PM, 30 August 2022)

The moving averages can be used for -----

Select the correct option

☐	Middle value,
☐	mode
☒	forecasting purposes
☐	Median

If in regression line Y on X the y intercept is zero then

Select the correct option



The line is parallel to x -axis



The line is parallel to y -axis



Y is multiple of X



None of these

Question # 3 of 10 (start time: 05:22:56 PM, 30 August 2022)

The points in scatter diagram show

Select the correct option

☐ No tendency to cluster around a straight line

☐ always relation between two variables

☐ None

☐ tendency to cluster around a straight line



Question # 6 of 10 (Start time: 07:38:59 PM, 30 August 2022)

In how many ways can the letters of the word "LEADER" be arranged?

Select the correct option

- | | |
|-----------------------|-----|
| ☐ | 720 |
| ☐ | 144 |
| ☐ | 72 |
| ☐ | 360 |

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When two events are mutually exclusive, the probability of either one of those occurring is the sum of individual probabilities. This is the called the ____

Select the correct option

☒	OR Rule
☐	AND-OR Rule
☐	AND Rule
☐	OR-AND Rule

Click to Save Answer & Move to Next Question

Question # 9 of 10 (start time: 05:33:21 PM, 30 August 2022)

State the addition theorem of probability for any two mutually exclusive events.

Select the correct option

☐ $P(A \text{ or } B) = P(A) - P(B)$

☐ $P(A \text{ or } B) = P(A) - P(B) + P(A \text{ and } B)$

☐ None of these

☒ $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$

Which of the following is not an example of seasonal variation?

Select the correct option

☐	annual earnings for a large corporation
☐	sales of snow blowers
☐	plant and flower sales
☐	use of electricity

Question # 1 of 10 (Start time: 07:31:48 PM, 30 August 2022)

When two events are mutually exclusive, the probability of either one of those occurring is the sum of the ____

Select the correct option

☐	OR-AND Rule
☐	AND Rule
☒	OR Rule
☐	AND-OR Rule

The sum of deviation of the observed values of Y_i from the regression line is always equal to

Select the correct option

☐

-1

☐

1

☐

None

☒

0



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$nCn = \text{-----}$

Select the correct option

☐ $nC0$

☐ 2

☐ None of these

☐ 3

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Question # 10 of 10 (start time: 05:16:03 PM, 30 August 2022)

A moving average is one of a family of similar techniques used to analyze

Select the correct option

☐	Regression model
☒	Time series data
☐	Correlation data
☐	Fitting a curve

Question # 8 of 10 (Start time: 10:53:52 PM, 30 August 2022)

If the equation of regression line is $y = 5$, then what result will you take out from it?

Select the correct option

☒

The line is parallel to x-axis.

☐

The line is parallel to y-axis.

☐

The line passes through (5, 0)

☐

The line passes through origin.

Question # 2 of 10 (start time: 05:03:17 PM, 30 August 2022)

The moving averages represent -----

Select the correct option

☒	Time series variations
☐	Statistical Dispersion
☐	Co-efficient of variations
☐	Absolute deviation

The moving averages of the Prices 90,70,30,110 are

Select the correct option



73.33, 80



63.33, 70



45.45, 68



65.50, 75

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The New Forecast is (constant of exponential smothing is 'c')

Select the correct option

☐	old forecast - c x (old actual + old forecast)
☒	old forecast + c x (old actual - old forecast)
☐	old forecast + c x (old forecast - old actual)
☐	None of these

Question # 10 of 10 (Start time: 07:44:38 PM, 30 August 2022)

Forecasting is a process in which

Select the correct option

☐	Assessment of a magnitude of a time series variable at some previous point of time
☐	The cyclically movements are assessed
☐	Assessment of a magnitude of a time series variable at some future point of time
☐	None of these

Question # 4 of 10 (Start time: 07:26:05 PM, 30 August 2022)

Probability of A and B happening = ____

Select the correct option

☐ Probability of A - Probability of B☐ Probability of A x Probability of B☐ Probability of A / Probability of B☐ Probability of A + Probability of B

What variation does moving average method eliminate?

Select the correct option

☐	Seasonal
☐	Cyclical
☐	Irregular
☐	Secular trend

If 'r' is the number of objects chosen, 'n' is the total number of objects and 'P' for permutation, Then which of the following is the formula for permutation?

Select the correct option



Reload Math Equations



$${}_nP_r = \frac{n!}{(n-r)!}$$



$${}_nP_r = \frac{r!}{n!}$$



$${}_nP_r = \frac{(n-r)!}{n!}$$



$${}_nP_r = (n-r)!$$

Click to Save Answer & Move to Next Question

The Marks of mathematics on weekly basis for three months

Period	Marks	Moving Averages
Month 1 Week 1	70	
Week 2	60	?
Week 3	65	?
Month 2 Week 1	80	?

Select the correct option

- ☐ 65
68.3
75.7
82.3
- ☐ 65
70
75.7
82.3
- ☐ 65
71
73
82.3
- ☐ 65
68.3
70
72
- 

Question # 6 of 10 (start time: 05:09:36 PM, 30 August 2022)

Pr of A (does not happen) = _____

Select the correct option

☐ None of these☐ $1 + \text{Pr of A (happens)}$ ☒ $2 \times \text{Pr of A (happens)}$ ☐ $1 - \text{Pr of A (happens)}$ 

The moving averages of the Prices 90,70,30,110 are

Select the correct option

☐ 73.33, 80

☐ 63.33, 70

☐ 45.45, 68

☐ 68.50, 75

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Question # 2 of 10 (start time: 05:21:27 PM, 30 August 2022)

The correct formula for OR rule is

Select the correct option

☒ $P(A \text{ or } B) = P(A) + P(B)$

☐ $P(A \text{ or } B) = P(A) - P(B)$

☐ None of these

☐ $P(A \text{ or } B) = P(A) P(B)$

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The moving averages can be used for

Select the correct option

☒	forecasting purpose ✓
☐	better graph
☐	middle value
☐	Median

An arrangement of all or some of a set of objects in a definite order is called _____

Select the correct option

☐	Factorial
☐	Forecast
☐	Combination
☐	Permutation



Question # 8 of 10 (Start time: 05:31:39 PM, 30 August 2022)

There are 5 white balls and 4 black balls. In how many ways can we select 3 white and 2 black balls?

Select the correct option

60

50

40

30

Probability of A and B happening = __

Select the correct option

☒	Probability of A + Probability of B	
☐	Probability of A - Probability of B	
☐	Probability of A x Probability of B	
☐	Probability of A / Probability of B	

Question # 7 of 10 (Start time: 07:40:19 PM, 30 August 2022)

A pattern of variation of a time series that repeats every year is called:

Select the correct option

☐	Secular
☐	Trend
☐	Seasonal
☐	Cyclical

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Question # 1 of 10 (Start time: 05:02:07 PM, 30 August 2022)

The moving averages of the Prices 20, 30, 40, 50, 60 are

Select the correct option

☐	20, 40, 60
☒	30, 40, 50
☐	40, 40, 40
☐	35, 45, 55

Question # 6 of 10 (**Start time: 10:44:42 PM, 30 August 2022**)

In tossing of a coin, the outcomes on successive trials are independent.

Select the correct option

True



False



Question # 2 of 10 (Start time: 07:33:29 PM, 30 August 2022)

The graph of time series is called .

Select the correct option

☐	Parabola
☐	Smooth Curve
☐	Histogram
☐	Straight line

Click to View Answer & Solution

The set of collection of all possible outcomes of an experiment is called

Select the correct option

☒	Sample Space
☐	None of these
☐	Sample set
☐	Power set

Question # 5 of 10 (Start time: 05:26:43 PM, 30 August 2022)

When a coin is tossed once, the probability of getting head is ...

Select the correct option

☐ 0.51

☐ 0.52

☒ 0.50

☐ 0.55

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The moving averages of the Prices 20, 30, 40, 50, 60 are

Select the correct option

☐	20, 40, 60
☐	30, 40, 50
☐	35, 45, 55
☐	40, 40, 40

Question # 3 of 10 (Start time: 05:05:00 PM, 30 August 2022)

The number of terms in the expansion of the binomial $(p + q)^n$ is

Select the correct option



$n+1$



n



$2n$



$n-1$

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Question # 1 of 10 (Start time: 05:19:55 PM, 30 August 2022)

Prices are 100, 110, 120, 130 and Moving averages are 110, 120
Now Actual- Trend are

Select the correct option

☐ 110, 120

☐ 10, 20

☐ 0, 0

☐ -10, 10



Question # 9 of 10 (Start time: 10:55:47 PM, 30 August 2022)

Which of the following is true for the special case $0!$ (zero factorial)?

Select the correct option



$0! = 1$



$0! = 10$



$0! = 100$



$0! = 0$

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Question # 6 of 10 (**start time: 05:28:33 PM, 30 August 2022**)

An arrangement of data by successive time periods is called

Select the correct option

☐ Representation of data

☐ Exponential smoothing

☒ Time Series

☐ Linear regression



Question # 1 of 10 (**Start time: 10:42:20 PM, 30 August 2022**)Evaluate $30! / 28!$. (! means factorial)

Select the correct option

☐

850

☐

28

☐

30

☐

870



If actual value is 180 and trend is 150, the seasonal variation is

Select the correct option

☐	40
☐	-30
☐	30
☐	-45

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The moving averages of the Prices 50, 60, 70, 80 are

Select the correct option



60, 70



65, 65



70, 80



50, 60

The correct formula for AND rule is

Select the correct option



$$P(A \text{ and } B) = P(A) + P(B)$$



$$P(A \text{ and } B) = P(A) - P(B)$$



None of these

$$P(A \text{ and } B) = P(A) P(B)$$

Question # 9 of 10 (Start time: 07:43:14 PM, 30 August 2022)

Which is the proportion of total variation explained by the regression line?

Select the correct option

☐	Standard error of the regression coefficient
☒	Simple coefficient of determination
☐	Slope of the regression line
☐	Mean square error

Question # 5 of 10 (Start time: 03:18:58 PM, 29 August 2022)

The moving averages of the Prices 50, 60, 70, 80 are

Select the correct option

☐ 65, 65☐ 70, 80☐ 50, 60☐ 60, 70

Question # 5 of 10 (Start time: 07:37:40 PM, 30 August 2022)

If a dice is thrown then the probability of getting an odd number or an even number is

Select the correct option

☐	None of these
☐	1
☐	$1/2$
☐	$1/3$



The following formula for probability

$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$ is called

Select the correct option

☐	The law of subtraction
☐	None of these
☐	the law of multiplication
☐	The law of addition

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In how many ways a team of 4 players be chosen from a total 10 persons?

Select the correct option

☐ 40

☐ 5040

☐ None of these

☒ 210



Question # 7 of 10 (Start time: 05:30:08 PM, 30 August 2022)

Which method of trend analysis is useful for data not having a pronounced trend or seasonality?

Select the correct option



Decomposition model



Multiplicative model



Exponential smoothing method



Trend analysis



In how many ways can a cricket eleven be chosen out of a batch of seventeen players?

Select the correct option



17C11



11C11



17C4



11C17

Prices are 2,4,6,8,10 and Moving averages are 4,6,8
then Actual- Trend are

Select the correct option



1, 2, 3



0, 0, 0



3, 4, 5



3, 2, 1

Question # 5 of 10 (start time: 05:08:24 PM, 30 August 2022)

A pattern of variation of a time series that repeats every year is called:

Select the correct option

☐	Secular
☒	Seasonal
☐	Cyclical
☐	Trend

If $A = \{1, 2, 3, 4\}$, $B = \{2, 4, 6\}$ then OR rule can be used to calculate $P(A \text{ or } B)$

Select the correct option

☐ False

☒ True

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Question # 9 of 10 (Start time: 05:14:05 PM, 30 August 2022)

The moving averages of the Prices 30, 45, 90, 110 are

Select the correct option

☐ 60, 85.45

☐ 65, 78.45

☐ 55, 81.67

☐ 45, 88.36

The correct formula for OR rule is

Select the correct option

☐	None of these
☐	$P(A \text{ or } B) = P(A) - P(B)$
☒	$P(A \text{ or } B) = P(A) + P(B)$
☐	$P(A \text{ or } B) = P(A) P(B)$

Question # 4 of 10 (Start time: 05:25:05 PM, 30 August 2022)

If $A = \{1, 2, 3, 4\}$, $B = \{2, 4, 6\}$ then OR rule can be used to calculate $P(A \text{ or } B)$

Select the correct option

☐ False

☒ True

The number of terms in the expansion of the binomial $(p + q)^n$ is

Select the correct option

☐	$n+1$
☐	$n-1$
☐	n
☐	$2n$

The sum of deviation of the observed values of Y_i from the regression line is always equal to

Select the correct option

☐ 1

☐ -1

☐ None

☒ 0

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When a pair of dice is thrown, the probability of getting a total of 5 or 11 is:

Select the correct option:

☒ $1/6$

☐ $1/7$

☐ $1/8$

☐ $1/5$



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Question # 10 of 10 (**Start time: 05:52:24 PM, 30 August 2022**)

When a straight line is fitted to time series data, it is called

Select the correct option

☒	Linear trend
☐	Linear equation
☐	Non-Linear equation
☐	Linear regression

Which of the following is correct

Select the correct option

☐	Random = Actual + Expected
☐	Random = Actual - Expected
☐	None of these
☐	Random = Expected - Actual



Question # 7 of 10 (start time: 05:11:36 PM, 30 August 2022)

Evaluate $13C_4$:

Select the correct option

815



715



615



None of these



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The graph of a regression line may be

Select the correct option



All of these



Linear



Cubic



Quardatic



When a straight line is fitted to time series data, it is called

Select the correct option

☐	Linear regression
☐	Linear equation
☒	Linear trend
☐	Non-Linear equation

Question # 1 of 10 (Start time: 10:38:03 PM, 30 August 2022)

When a coin is tossed, the events that it lands heads or tails are mutually exclusive

Select the correct option

True



False



Question # 10 of 10 (Start time: 05:35:16 PM, 30 August 2022)

Forecasting is a process in which

Select the correct option



Assessment of a magnitude of a time series variable at some previous point of time



Assessment of a magnitude of a time series variable at some future point of time



The cyclically movements are assessed



None of these

Question # 4 of 10 (Start time: 05:06:55 PM, 30 August 2022)

Which of the following values generally perform best for smoothing constant ' α ' (alpha)?

Select the correct option

☒	' α ' between 0.1 and 0.3
☐	' α ' between 1 and 3
☐	' α ' between 0 and 1
☐	' α ' between 0.5 and 1

Question # 3 of 10 (**Start time: 10:47:14 PM, 30 August 2022**)

In tossing of a coin, the outcomes on successive trials are independent

Select the correct option

☐	False
☒	True

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